# ESE 123 Lab 3 Report: Voltage Divider

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**Objective.** Understand the voltage divider formula, check it with an LTspice simulation, and use it to find the test-point voltage TP1 on the project board.

## 1. The voltage divider formula

$V_{in}$ drives $R_{top}$ and $R_{bottom}$ in series, and $V_{out}$ is taken across $R_{bottom}$. Both resistors carry the same current I:

> $I = V_{in} / (R_{top} + R_{bottom})$ (series resistors, Ohm's law)
> $V_{out} = I \cdot R_{bottom} \longrightarrow V_{out} = V_{in} \cdot R_{bottom} / (R_{top} + R_{bottom})$

Voltage is shared in proportion to resistance, so $V_{out}$ is always between 0 and $V_{in}$. The formula assumes nothing else draws current from the middle node. If a load is connected at $V_{out}$, it must be combined in parallel with $R_{bottom}$ first.

## 2. LTspice simulation (V1 = 3 V, R1 = 1 kΩ, R2 = 2 kΩ)

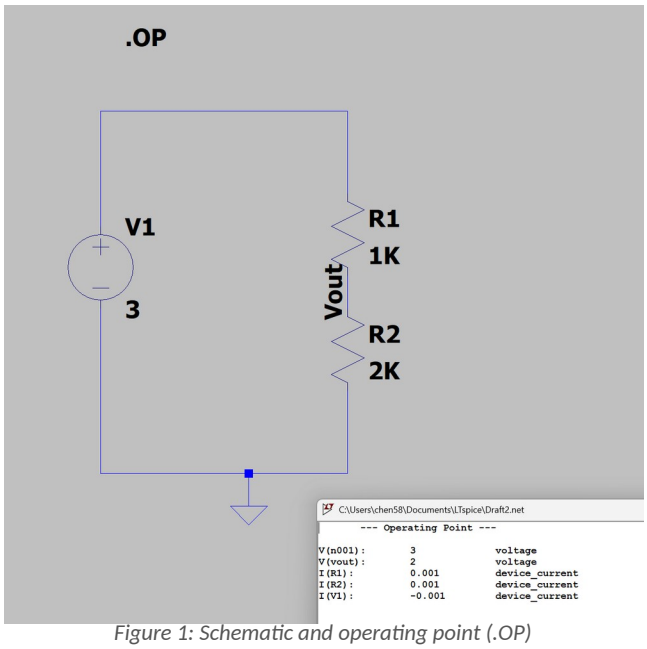



*Figure 1: Schematic and operating point (.OP)*

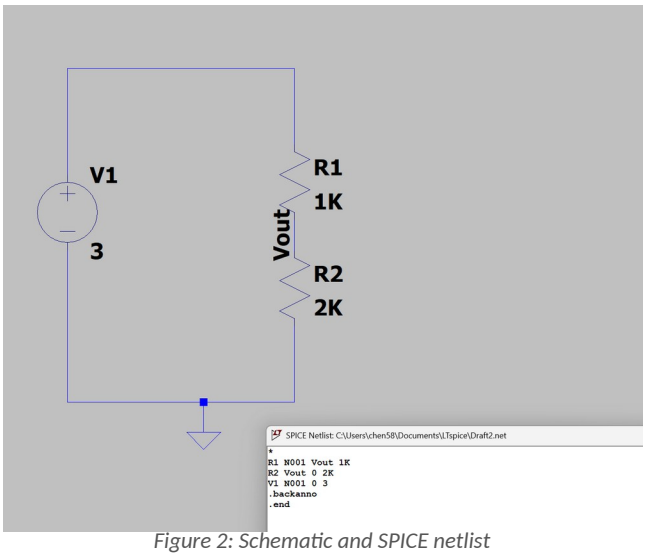



*Figure 2: Schematic and SPICE netlist*

| Quantity | Hand calculation | LTspice (.OP) |
|---|---|---|
| $V_{out}$ | 3 · 2k / (1k + 2k) = 2 V | 2 V |
| Current I | 3 V / 3 kΩ = 1 mA | 0.001 A |

The simulation matches the hand calculation exactly. R2 is twice R1, so it takes two thirds of the 3 V supply (2 V), and the same 1 mA flows through both resistors.

## 3. Project board divider (TP1)

The board has two resistors on top (R1, R2) and three on the bottom (R3, R4, R5), fed from 5 V. Each group is collapsed into one equivalent resistor. Then the divider formula is applied.

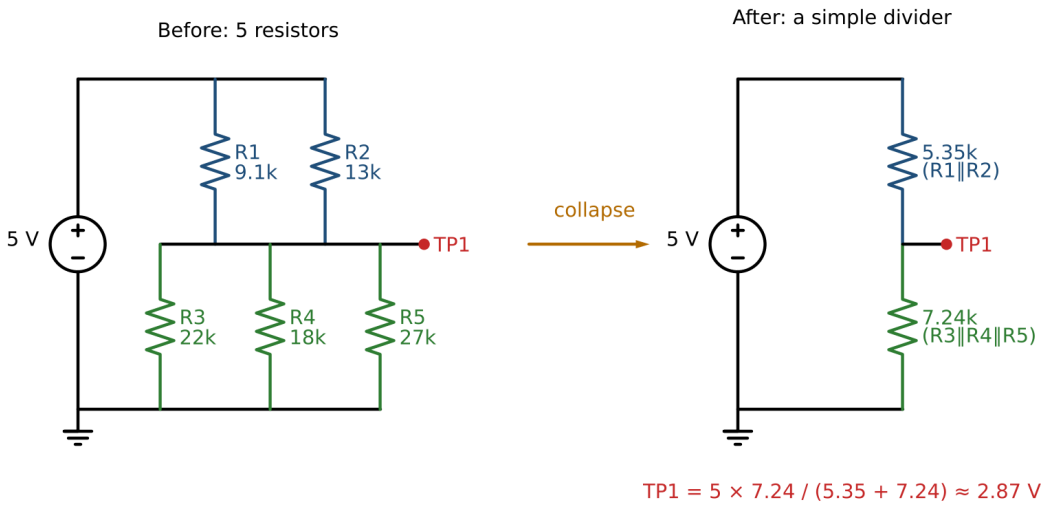



*Figure 3: Collapsing the board circuit into a simple divider*

| Step | Calculation | Result |
|---|---|---|
| Top group | R1 ∥ R2 = 9.1k · 13k / (9.1k + 13k) | 5.35 kΩ |
| Bottom group | 1 / (1/22k + 1/18k + 1/27k) | 7.24 kΩ |
| Divider | 5 V · 7.24k / (5.35k + 7.24k) | **TP1 ≈ 2.87 V** |

Check: a parallel combination is always smaller than its smallest resistor (5.35k < 9.1k, 7.24k < 18k). TP1 is between 0 and 5 V, as it must be.